

Shubh Gupta

+91 9407515878 | shubhgupta874@gmail.com | linkedin.com/in/sh10bh | github.com/Sh10bh | Bhopal, India

Summary

3rd-year Computer Science undergraduate at VIT Bhopal with hands-on experience building and deploying end-to-end AI/ML systems — from LLM-powered pipelines and RAG applications to predictive ML models. Seeking ML/backend internships to apply expertise in applied AI and production-grade deployment.

Education

VIT Bhopal University

Bachelor of Technology in Computer Science | CGPA: 8.09

Bhopal, India

Sept. 2023 – May 2027

Technical Skills

Languages: Python, JavaScript, C/C++, HTML/CSS

AI & LLM: LangChain, ChromaDB, HuggingFace Embeddings, Groq API (Llama 3.3-70B), YOLOv8, Pydantic v2

ML & Data Science: PyTorch, Scikit-learn, XGBoost, LightGBM, SMOTE, Pandas, NumPy, Matplotlib

Backend & DevOps: FastAPI, Docker, Docker Compose, GitHub Actions, Git, Linux

Frontend & Deployment: React, Streamlit, Vercel, HuggingFace Spaces, Render, Streamlit Cloud

Projects

Forge — LLM App Config Generator | Python, FastAPI, Groq API, Pydantic v2 | GitHub | Live

- Architected a **4-stage LLM pipeline** (Intent → Design → Schema → Repair) converting natural language prompts into fully validated JSON app configs spanning DB, API, UI, and Auth layers.
- Engineered strict **Pydantic v2 schema validation** with a targeted repair engine classifying 3 failure types and surgically fixing only the broken layer, achieving **83.3% first-pass JSON success** across 12 diverse prompts.
- Sustained **600k tokens/day** on the free tier via 6-key Groq API round-robin rotation; deployed on Render with auto-deploy CI/CD via render.yaml.

AskMyPDF — RAG Document Q&A System | Python, FastAPI, LangChain, ChromaDB, React | GitHub | Live

- Built a full-stack **Retrieval-Augmented Generation** app for natural language PDF querying with source citations; semantic retrieval via ChromaDB vector store and HuggingFace embeddings (all-MiniLM-L6-v2).
- Implemented persistent multi-turn conversation memory via LangChain; containerised with Docker Compose, deployed backend on HuggingFace Spaces and React frontend on Vercel with CI/CD via GitHub Actions.

Customer Churn Prediction | Python, LightGBM, Scikit-learn, SMOTE | GitHub | Live

- Engineered an end-to-end ML pipeline for telecom churn prediction; resolved class imbalance with SMOTE and built a ColumnTransformer preprocessing pipeline.
- Tuned LightGBM via GridSearchCV, achieving **ROC-AUC of 0.83**; deployed as a live Streamlit web app for real-time churn probability scoring.

Delhi AQI Prediction Model | *Python, XGBoost, Scikit-learn* | GitHub | Live

- Built an XGBoost regression model on Delhi air quality data with GridSearchCV hyperparameter tuning, resolving critical overfitting and achieving **R² of 0.93+**; deployed as a live Streamlit app.

Research & Preprints

Real-Time Data Optimisation for Object Detection via Information Theoretic Network Coding | SSRN, 2025

- Fused **YOLOv8** real-time detection with **Sparse RLNC** in a smart traffic system, cutting intersection wait times by **26.9%** and packet loss from 12.4% to **3.9%** under congested network conditions.

Improving Object Detection Efficiency via Joint Information Theory and Network Coding | SSRN, 2025

- Proposed a joint information-theoretic coding framework for object detection, achieving **9.03%** accuracy improvement and **7.48%** reduction in packet loss under simulated network congestion.

Positions of Responsibility

Vice-Captain & Club Co-ordinator, Sports Club (Football)

VIT Bhopal University

April 2024 – Present

Bhopal, India

- Led logistics for 4 major college fests (1000+ attendees); achieved Runner-up at AdvITYa 2025 and 4th place at SPARDHA 2024 (IIT BHU).